



Andrej Karpathy

I like to train deep neural nets on large datasets 🧠 🤖 ⚡



2024 -



I create educational videos on AI on my [YouTube channel](#). The videos come in two parallel tracks: a technical track and a general audience track.

1. Technical track: Follow the [Zero to Hero](#) playlist. General audience track:
2. [Deep Dive into LLMs like ChatGPT](#) is on under-the hood fundamentals of LLMs.
3. [How I use LLMs](#) is a more practical guide to examples of use in my own life.
4. [Intro to Large Language Models](#) is a third, parallel, video from a longer time ago.

For all the latest, I spend most of my time on [X/Twitter](#) or [GitHub](#).

2023 -
2024



I came back to [OpenAI](#) where I built a new team working on midtraining and synthetic data generation.

2017 -
2022



I was the [Director of AI at Tesla](#), where I led the computer vision team of [Tesla Autopilot](#) and (very briefly) [Tesla Optimus](#). My team handled all in-house data labeling, neural network training and deployment on Tesla's custom inference chip. Today, the Autopilot increases the safety and convenience of driving, but the team's goal is to make [Full Self-Driving](#) a reality at scale. See Aug 2021 [Tesla AI Day](#) for more.

2015 -
2017



I was a research scientist and a founding member at [OpenAI](#).

2011 -
2015



My PhD was focused on convolutional/recurrent neural networks and their applications in computer vision, natural language processing and their intersection. My adviser was [Fei-Fei Li](#) at the Stanford Vision Lab and I also had the pleasure to work with [Daphne Koller](#), [Andrew Ng](#), [Sebastian Thrun](#) and [Vladlen Koltun](#) along the way during the first year rotation program.

I designed and was the primary instructor for the first deep learning class Stanford - [CS 231n: Convolutional Neural Networks for Visual Recognition](#). The class became one of the largest at Stanford and has grown from 150 enrolled in 2015 to 330 students in 2016, and 750 students in 2017.

Along the way I squeezed in 3 internships at (baby) Google Brain in 2011 working on learning-scale unsupervised learning from videos, then again in Google Research in 2013 working on large-scale supervised learning on YouTube videos, and finally at DeepMind in 2015 working on the deep reinforcement learning team with [Koray Kavukcuoglu](#) and [Vlad Mnih](#).

2009 -
2011



MSc at the University of British Columbia where I worked with [Michiel van de Panne](#) on learning controllers for physically-simulated figures (i.e., machine-learning for agile robotics but in a physical simulation).

2005 -
2009



BSc at the University of Toronto with a double major in computer science and physics and a minor in math. This is where I first got into deep learning, attending [Geoff Hinton's](#) class and reading groups.

bio

Andrej Karpathy is an AI researcher and educator. He was a founding member of OpenAI and later the Director of AI at Tesla, where he led the computer vision team of the Autopilot. During his PhD at Stanford he was the architect and lead instructor of the first deep learning course at Stanford (CS231n), which has become one of its most popular classes.

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